

Quick Check

Lesson 7: Putting the Evidence Together: Why Cassava Cuttings Need No Seeds but Lone Pawpaw and Avocado Trees Need a Partner

8 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.1: Reproduction in Plants | Lesson 7

A farmer in Kisumu plants a piece of cassava stem in the soil. It grows into a whole new plant without any flower, pollen or seed. Which statement best explains this?

- A** The stem cutting must first be pollinated by insects before it can grow
- B** The cutting grows only because it has a flower that is too small to see
- C** New roots and shoots grow from the stem tissue of the parent, so no flowers or fertilisation are needed
- D** The cutting is really a seed that is hidden inside the stem

In a hibiscus flower, the anthers release their pollen before the stigma of the same flower is ready to receive it. What is this feature called, and what does it do?

- A** Dioecy: male and female flowers are on separate plants
- B** Protandry: the timing difference makes self-pollination less likely
- C** Monoecy: male and female flowers are on the same plant
- D** Herkogamy: the stigma and anthers are at different heights

A maize plant has male tassels at the top and female cobs with silk lower down on the same plant. Which description is correct?

- A** It is protandrous, because the cob grows from the tassel
- B** It is self-incompatible, because it has no male parts
- C** It is monoecious, because separate male and female flowers grow on the same plant
- D** It is dioecious, because it has male and female parts on separate plants

During double fertilisation in a flowering plant, two sperm nuclei from the pollen tube enter the ovule. What does each one do?

- A** One forms the seed coat and the other forms the fruit wall
- B** Both fuse with the egg to form two embryos
- C** One fuses with the egg to form the endosperm, and the other fuses with the polar nuclei to form the zygote
- D** One fuses with the egg to form the zygote, and the other fuses with the polar nuclei to form the endosperm

A bean pod is opened and the seeds are found attached in a row along one edge of the pod. What type of placentation is this?

A Parietal placentation

B Free-central placentation

C Axile placentation

D Marginal placentation

After pollination and fertilisation in a tomato flower, which part of the flower develops into the fruit and which part develops into the seeds?

- A** The ovary becomes the fruit and the ovules become the seeds
- B** The ovules become the fruit and the ovary becomes the seeds
- C** The petals become the fruit and the anthers become the seeds
- D** The stigma becomes the fruit and the style becomes the seeds

A farmer in Kericho plants a whole field with cassava cuttings taken from one parent plant. A disease that attacks this variety arrives. Why is the whole field at high risk?

- A** Cuttings produce seeds that carry the disease
- B** Cuttings from one parent are genetically identical, so they all lack resistance to the disease
- C** Cuttings need pollinators, and the disease kills them
- D** Cuttings always grow more slowly than seeds, so they are weak

A lone avocado tree in a compound in Kiambu is covered in flowers every year but never sets fruit. Which explanation best uses what we know about barriers to self-pollination?

- A** Flowers are only for decoration, so they never lead to fruit
- B** Avocado trees can only be grown from cuttings, so they cannot form fruit
- C** Its flowers cannot be fertilised by its own pollen (self-incompatibility), so it needs pollen from another avocado tree
- D** The tree has too many petals, which stop the ovary from growing

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).